UNIT-3 INDUSTRIAL PHARMACY

ONESHOT - CRASH COURSE

B PHARMACY 5TH SEMESTER

IMPORTANT QUESTIONS

M.M.M.I.

1. Define capsule and their types. Explain production (manufacturing/preparation) of capsule with process of filling:- Hard gelatin capsule shell
   Soft gelatin capsule (Rotary Die process). → packaging, storage, stability testing.

M.I.

2. Discuss the parameters of in-process and final products quality control tests for capsule.
3. Explain the formulation, requirement and process of pelletization for pharmaceutical pellets.

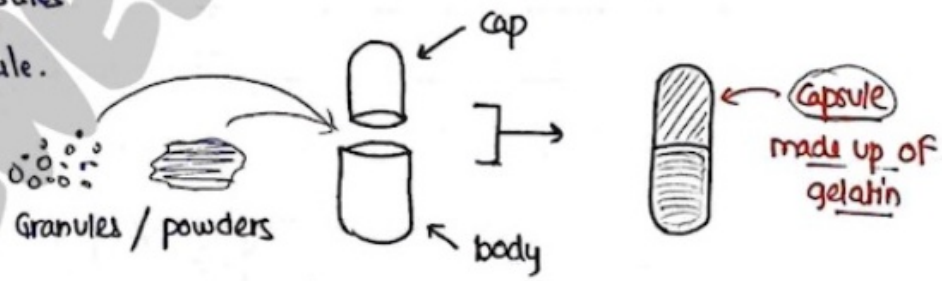
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CAPSULES

- Capsules are solid unit dosage forms in which a drug or mixture of drug is enclosed inside a small, easy-to-swallow shell.
- This shell is usually made of gelatin (hard or soft) but sometimes made from vegetables polymers for vegetarian use.
- Capsules protect the drug from the external environment, improve stability and help in accurate dosing.
- They are widely used because they offer better appearance, easy administration and can carry different types of materials like powders, granules, pellets and even liquids in soft capsule

  Eg. Amoxicillin capsules

  Omeprazole capsule.



Advantages -

- Capsules are easy to swallow due to their smooth and slippery gelatin shell.
- They effectively mask the bad taste and odour of drug by enclosing them completely.
- They protect drugs from air, light and moisture, improving overall stability.
- They can be filled with powders, granules, pellets, tablets and even liquids (in soft capsule).

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Disadvantages

- Capsules are moisture-sensitive and may become soft or brittle depending on humidity.
- Some drugs react with gelatin shells, leading to stability problems.
- Higly soluble salts may cause gastric irritation when filled in capsules.
- Soft gelatin capsules require costly materials and specialized equipment.
- Improve storage may cause microbial growth or deformation of the capsule shell.



Types of Capsules

Capsules are of two types:

1. Hard Gelatin Capsules - These are two piece capsules consisting of a body and a cap mainly used to fill dry materials like powders, granules and pellets.
   Eg. Amoxicillin capsule, Paracetemol capsule.

2. Soft Gelatin Capsules - These are one-piece, soft, elastic capsules used to fill liquid, oils, and semisolid drugs.
   Eg. Vitamin E soft gel, Omega-3- capsules.



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Hard Gelatin Capsule.

These are two-piece (solid) dosage forms made of a gelatin shell that consists of a "body" and a "cap", designed mainly for (filling) dry materials like powders, granules, pellet and small tablets.

- They are widely used in industries because they are easy to produce, cost-effective, provide accurate dosing and offer flexibility in filling different solid materials. The shell is usually made from (gelatin), water and colorants and it dissolves quickly in the stomach to release the drug.

Eg. Amoxicillin capsule.
Diclofenac sodium capsule.



(Size) of Hard Gelatin Capsules.

Capsules are manufactured in different standard (sizes) to hold different qualities of drug material, and these (sizes) range from very small (for potent drugs) to very large (for high-dose drugs).

Range of sizes - Hard gelatin capsules are available in size 000 to 5.

- 000 - Largest (size)
- 5 - Smallest (size.)



Commonly used sizes -

- Size 0, 1, 2 - Most commonly used for (adult) doses (200 - 800 mg)
- Size 4, 5 - Used for pediatric or low dose drugs (60 - 200 mg)
- Size 000 - Used for very high-dose formulation (800 - 1600 mg).

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Production of Hard Gelatin Capsule.

Q. Explain production of Hard Gelatin capsule shell with various filling methods.

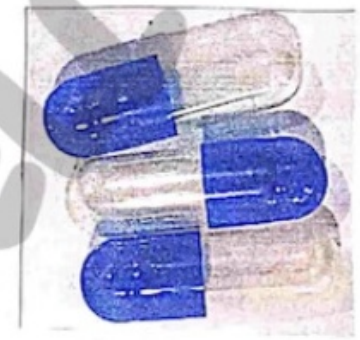
Hard gelatin capsules are prepared through two combined stages, first the empty capsule shell is manufactured from gelatin solution to form the body and cap, and then these empty shells are filled with powders, granules, pellets or tablets to produce a complete, ready to use dosage form.

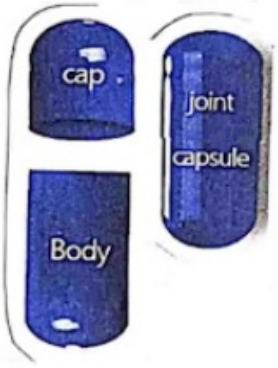


Formulation of Hard Gelatin Capsule Shell.

The capsule shell mainly contains the following ingredients:

(i) Gelatin
(ii) Water
(iii) Colorants / Dyes.
(iv) Opacifier.
(v) Preservatives
(vi) Plasticizers.





① **Gelatin** - Gelatin is a natural protein obtained by partial hydrolysis of Collagen present in animal bones, skin and connective tissues

- It forms a clear, strong, flexible film which makes it ideal for capsule shells.
- It is the main forming material used to prepare the capsule shell.
- It Provides strength, transparency, smoothness and film-forming ability to the shell.

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Types of Gelatin

1) Type A Gelatin (Acid-treated gelatin)

- Prepared by acid treatment of collagen. → Isoelectric point (pI) ↓ when gelatin charge is zero
- Has **isoelectric point pH 7-9** (slightly acidic environment).
- Produce soft, clear, flexible films.
- Used when quick processing is required.

Eg. Pig skin gelatin.



11) Type B (Alkali-treated gelatin).

- Prepared by alkali treatment of collagen. (slightly basic)
- Has **isoelectric point pH 4-5.**
- Produces stronger and more stable films.
- Used for stronger shells and longer production cycles.

Eg. Bovine (cow) bones
- Calf hides.





(iii) Water.

- Used to hydrate and dissolve gelatin
- Maintains proper viscosity of the gelatin solution during dipping
- Helps in film formation on pins during shell making.

(iii) Colorants / Dyes



- Provides identification and appearance
- Prevent mix-ups by using different colors for body and caps.

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(IV) Opacifiers

- Make the shell opaque and protect light-sensitive drugs.

Eg • Titanium dioxide.

(V) Preservatives

- Prevent microbial growth in gelatin mass because gelatin contains moisture.

Eg • Parabens.

(VI) Plasticizers.

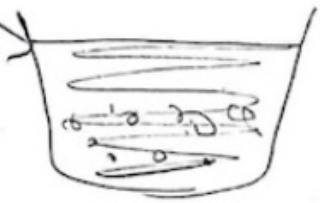
Eg • Improve flexibility of the shell and reduce brittleness.

Eg. Glycerin, Sorbitol. (mainly in soft gels).

Procedure For Manufacturing Hard Gelatin Capsule Shell.

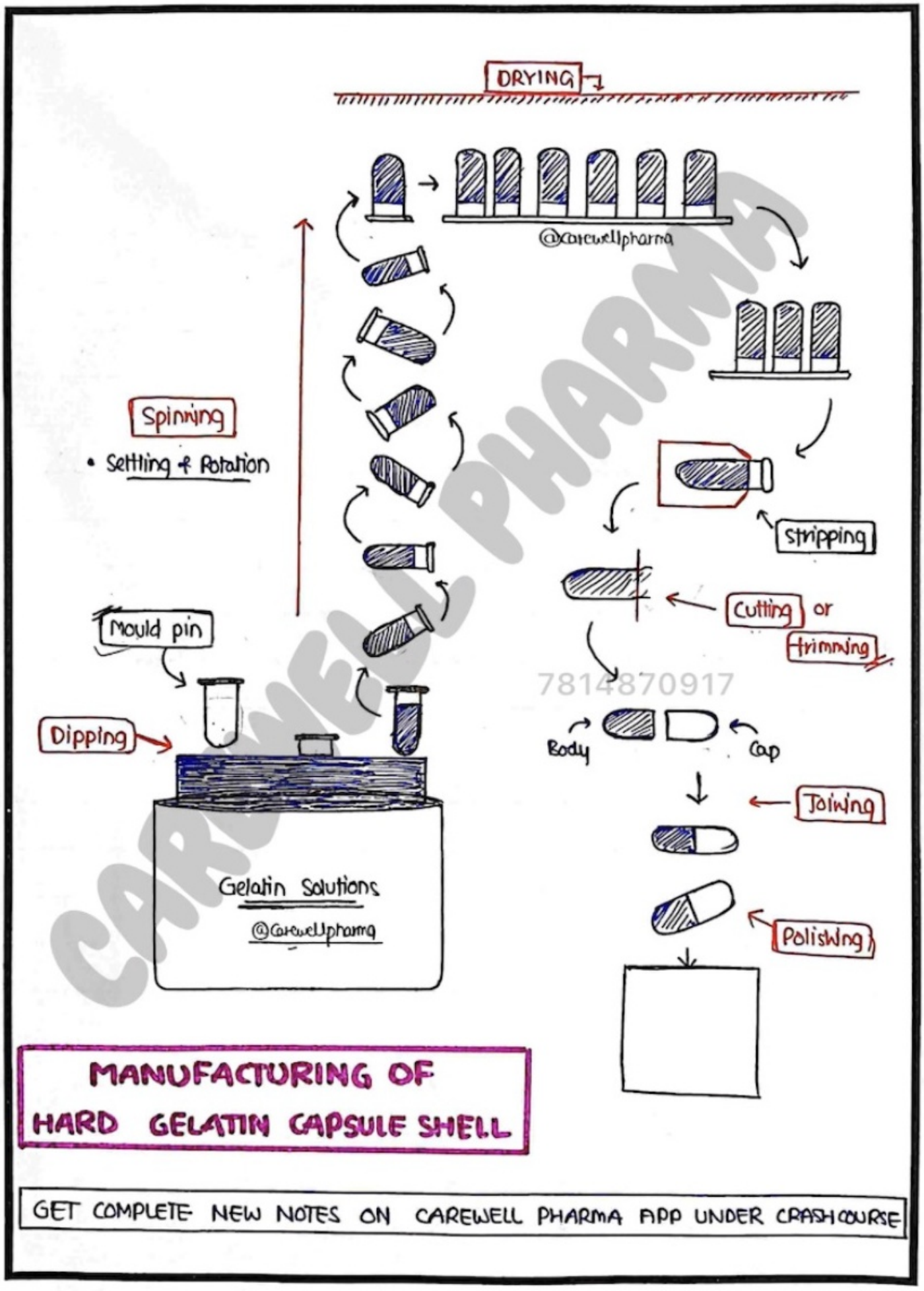
It involve several steps:-

1. Preparation of gelatin solution



2. Pin dipping.
3. Setting and rotation
4. Drying
5. Stripping and trimming
6. Joining and polishing.
7. Inspection, sorting and storage.

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GET COMPLETE NEW NOTES ON CAREWELL PHARMA APP UNDER CRASH COURSE

1. Preparation of Gelatin Solution.- The process begins by mixing gelatin, water and other ingredients to make a smooth liquid that will be used to form the capsule shell.

2. Pin Dipping - Once the gelatin solution is ready, stainless-steel pins are dipped into it, so that a thin film of gelatin sticks to the pins and begins the formation of the body and cap.

3. Setting and Rotation - After dipping, the coated pins move forward and rotate slowly, which helps the wet gelatin film spread evenly and start setting into a soft shell.

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4. Drying - As the gelatin film begins to set, the pins enter a drying chamber where warm air removes moisture and turns the soft film into a firm capsule shell.

5. Stripping and Trimming - When the shells become firm after drying, they are gently removed from the pins and then trimmed to the correct length to allow proper fitting of body and cap.

6. Joining and Polishing - The body and cap are partially joined to maintain shape, and the shells are polished to remove any dust and improve their smooth appearances.

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7. Inspection, Sorting and Storage - Finally, the polished shells are checked for defects, sorted and then stored in controlled conditions so they stay stable until the filling process.

M.M.I

Various Filling Process of Hard Gelatin Capsules.

Filling of hard gelatin capsules means transferring the drug material into the empty capsule body and then closing the capsule with its cap to form a complete, ready-to-use dosage form.

- This step is done after the capsule shell has been manufactured and involves dosing, filling, locking and finishing.
- It involves three methods :-

(i) Manual Filling Method - In this method, Capsules are filled by hand or by using a small hand-operated device.

Hand operated capsule filling machine

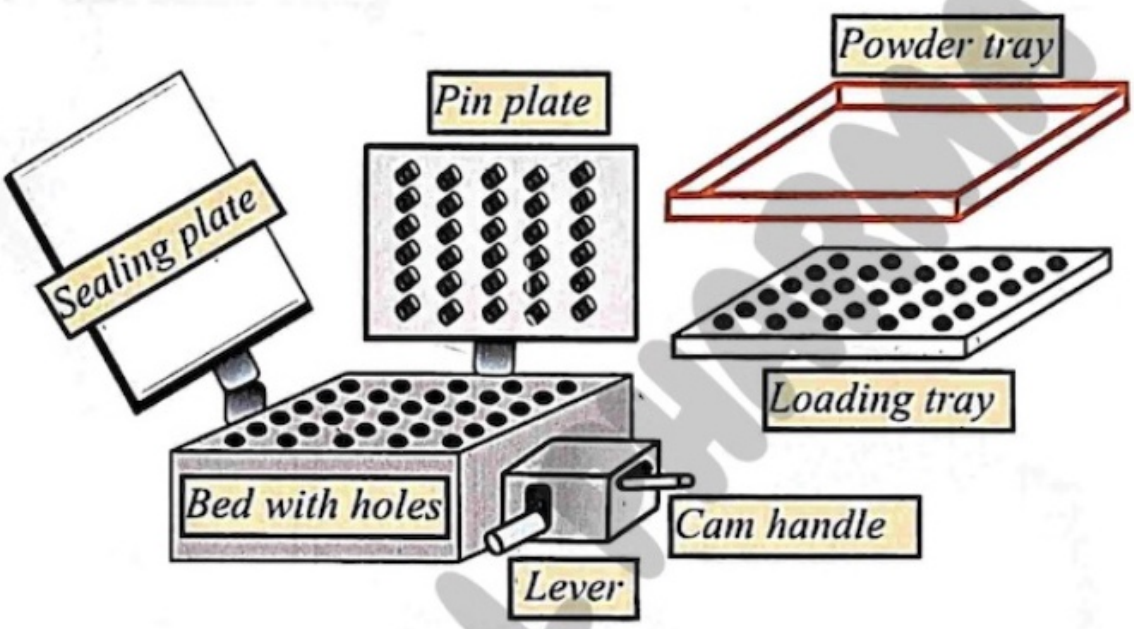
Procedure -

- The capsule cap and body are separated manually.
- The open body is pressed into a powder bed until it gets filled.
- Extra powder is removed.
- The cap is placed back and the capsule is locked.

- It is used for small batches and laboratory work.

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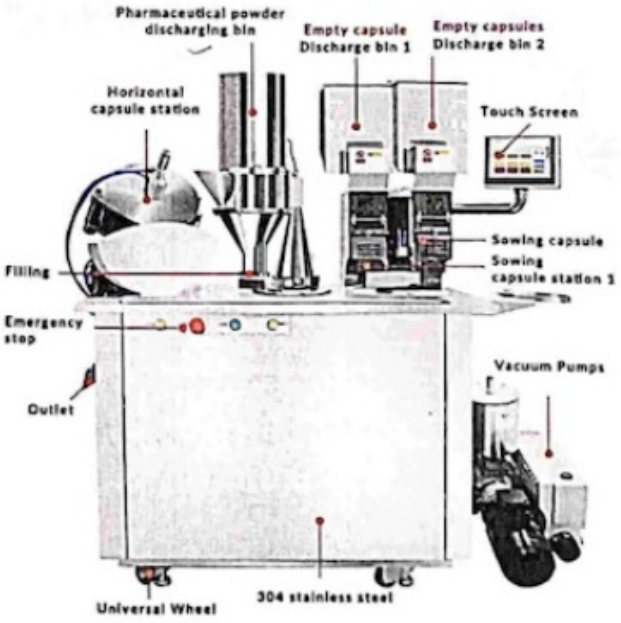
HAND OPERATED MACHINE



AUTOMATIC MACHINE



SEMI-AUTOMATIC MACHINE



2. Semi - Automatic filling Method - In this method, Some parts of the filling are done by the machine and some parts are done by the operator.

Procedure -

- The machine separates the cap and body automatically.
- Powder is filled into the open bodies using a dosing system.
- The powder is lightly pressed to make the dose accurate.
- The operator helps in closing and collecting the capsules.

It is used for Medium - scale manufacturing.

3. Fully Automatic filling Method - In this method, The entire filling process is done by the machine without manual handling.

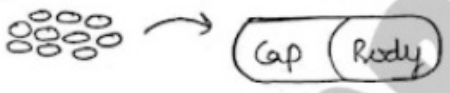
Procedure -

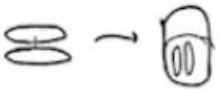
- The machine separates caps and bodies on its own.
- The machine fills the exact dose of powder into each body.
- Tamping pins adjust the powder level.
- The machine locks the cap onto the body automatically.
- finished capsules come out ready and polished.

It is used for Large - scale industrial production.

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Special Techniques of Hard Gelatin Capsule formulation.

(i) **Pellet Filling** – In this method, multiple-unit pellets prepared by extrusion-spheronization or coating are filled into the capsule body to create controlled or delayed release dosage forms. 

(ii) **Tablet in Capsule filling** – A small compressed tablet is placed inside the capsule body to combine two different doses or to achive modified drug release. 

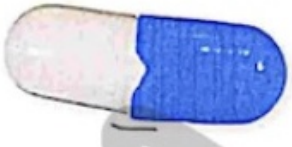
(iii) **Liquid or Semi-solid Filling (sealing after filling)** – Lipid-based liquids or 
semisolid drug systems are filled into hard gelatin capsules using special low-viscosity formulations and the capsule cap may be banded or sealed to prevent leakage.

(iv) **Combination filling** – This technique combines powders, pellets and tablets together in one capsule to achieve multi-dose or multi-release action.

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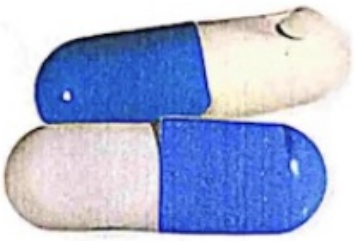
MANUFACTURING DEFECTS

JOINTS



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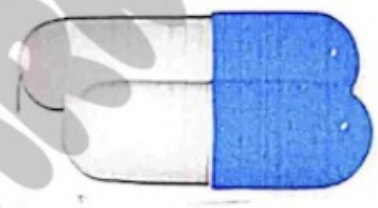
DENTS



TIP DEFECT



LENGTH



HOLES

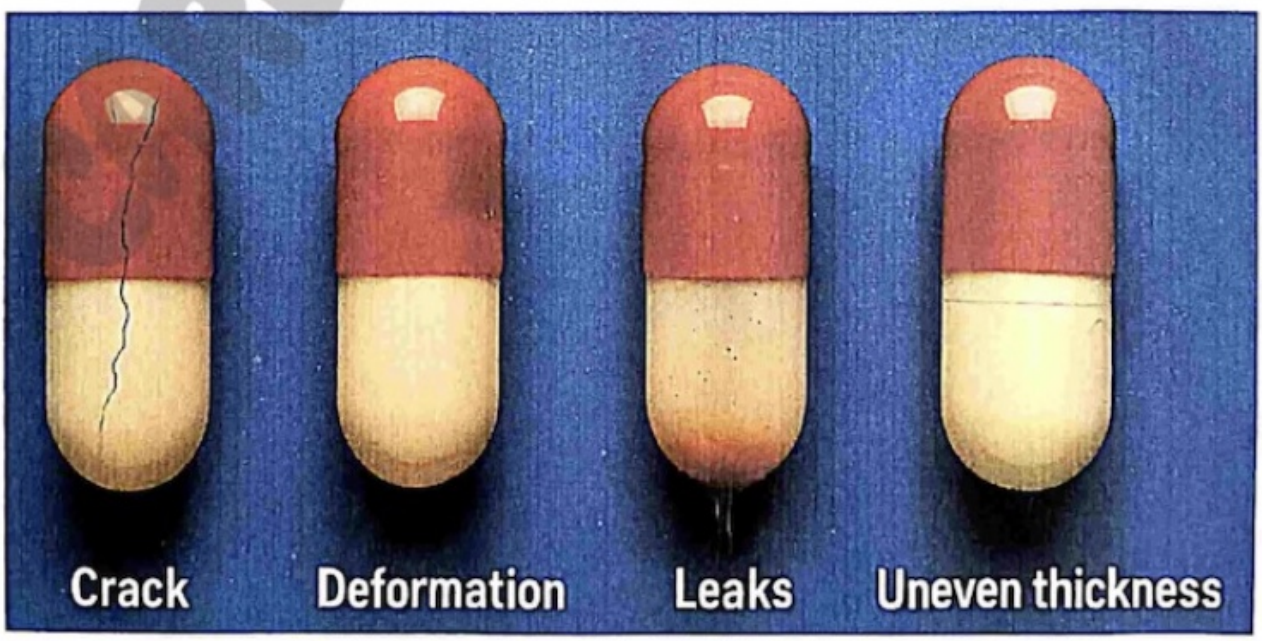


DOUBLE CAP



PRINT DEFECTS





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Manufacturing Defects of Hard Gelatin Capsules.

(i) **Dents or Deformed shells** - Capsule shells may get pressed or deformed due to improper handling or mechanical pressure during production.

(ii) **Cracks or Splits** - Small cracks develop on the body or cap when gelatin becomes too dry or brittle.

(iii) **Pinholes** - Tiny holes appear on the shell surface due to air, bubbles or dust particles during gelatin dipping.

(iv) **Color Variation** - Uneven color or light / dark patches occur due to poor mixing of colorants in the gelatin solution.

(v) **Improper Cap-Body fit** - The cap does not lock properly on the body due to wrong trimming, size mismatch, or deformation.

(vi) **Bubble formation** - Air bubbles get trapped inside the gelatin film during dipping, creating weak spots on the shell.

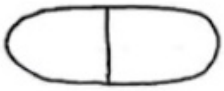
(vii) **Brittleness or Softening** - Capsules become too brittle in low humidity or too soft in high humidity due to moisture imbalances.

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(viii) Black specks / foregin Particles - Dust, fibres or foregin particles get embedded in the shell during production.

Quality Control Tests.

1. Appearance / Visual Inspection - This test checks the capsule for color uniformity, smooth surface, absence of dents, cracks, spots, leakage or deformation to ensure physical quality.



Procedure -

- Capsules are placed on a white background under good light.
- Inspector visually examines them or uses an automatic camera system.
- Defective capsules are separated.

2. Weight Variation Test - In this test, individual capsules are weight to confirm that each unit contains almost the same total weight and does not shows excessive variation. 20 capsules = fill capsule - empty capsule = Actual weight of drugs.

Procedure -

- Take 20 capsules and weigh each capsule (W_1)
- Empty them and weigh empty shells (W_2)
- Drug fill weight = $W_1 - W_2$.
- Compare variation with pharmacopoeial limits.

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3. **Content Uniformity Test** – This test measures the **amount** of active ingredients present in individual capsule to make sure each capsule contains accurate and uniform drug content.



Procedure

for each capsule

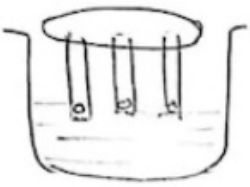
- Take 10 capsules.
- Open each **capsule** and dissolve content in suitable solvent.
- **Analyze** drug amount using **UV** / HPLC.
- Each capsule must lie within **85% – 115%** of label claim (as per IP/USP).

4. **Disintegration Test** – It checks the **time** required for the capsule shell to break and release its contents under standards conditions, ensuring **proper** breakdown in the digestive system.



Procedure –

- Place **capsules** in each tube of the disintegration apparatus.
- Use **water** or 0.1 N HCl to 37°C
- Move basket up and down
- Record time when all capsules break completely.



Conditions:	Medium	Temperature	Limit
(i) Hard-Gelatin Capsule	Water/buffer	37° ± 2°C	30 min or as per individual monograph
(ii) Soft-Gelatin Capsule	Water	37° ± 2°C	60 min or as per individual monograph

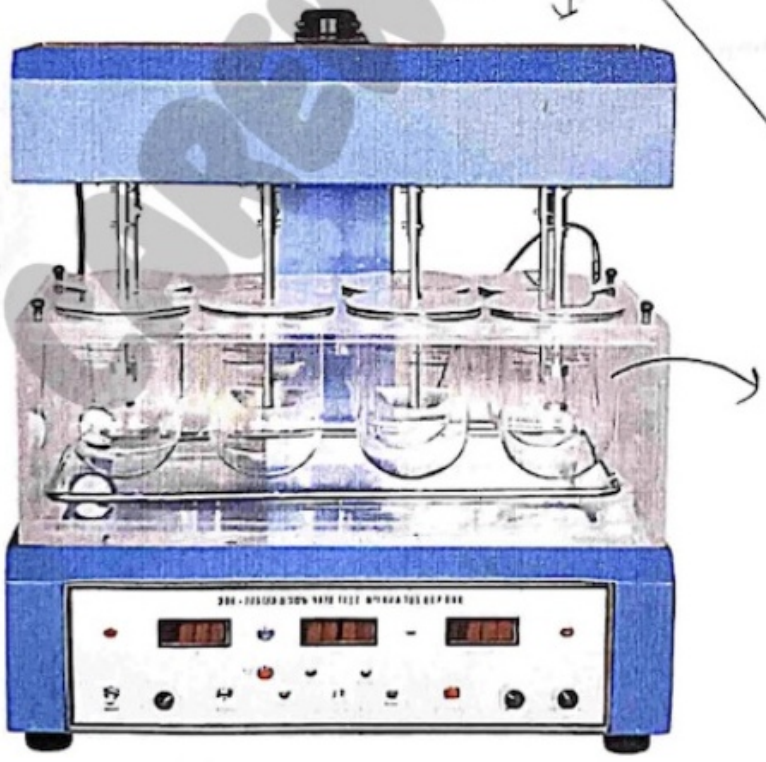
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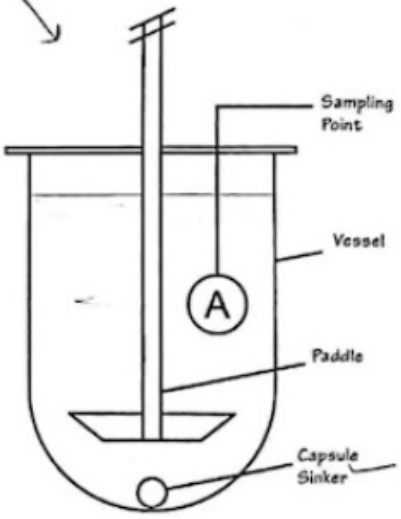
DISINTEGRATION APPARATUS



DISSOLUTION APPARATUS







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5. Dissolution Test - This test measures how fast and how much drug dissolves from the capsule in a given time, predicting drug release and absorption in the body.

Procedure -

- Use dissolution apparatus (Basket or Paddle)
- Capsule placed in medium (like 900 ml HCl)
- Sample drawn at fixed time intervals.
- Amount dissolved measured by UV/HPLC
- Compare with % release requirement.

6. Assay (Drug Content Test) - Assay determines the exact amount of active drug present in the capsule and compares it with the labeled claim for accuracy.

Procedure

- Powder content of several capsules.
- Dissolve in suitable solvent.
- Analyze using titration / UV/HPLC
- Compare with label claim (usually 98 - 102 %).



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Soft Gelatin Capsules.

These are one piece, sealed capsules made from soft, elastic gelatin, designed mainly for filling liquids, oils and semi-solid drug formulation.

- They are made by a special process called the Rotary Die Process, where the shell is formed, filled and sealed in a single operation.

Eg. Vitamin E softgel, Omega-3 fish oil capsules.



Single piece capsule

Liquid, oils and semi-solid drug formulation.

Formulation of Soft Gelatin Capsule shell.

Softgel shell contains the following ingredients:

(i) Gelatin
(ii) Plasticizers
(iii) Water
(iv) Colorants / opacifiers
(v) Preservatives
(vi) Flavoring agents.



Soft-gelatin Capsule (single-piece)

(i) Gelatin - It forms the main structure of the softgel shell. It gives strength, clarity and elasticity.

- Type A or Type B gelatin can be used.
- Eg. pig skin or cow bones.

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(ii) Plasticizers (Glycerin or sorbitol) - It makes the capsules shell soft, flexible and elastic.

- More plasticizer = softer shell.

Eg. Glycerin, Sorbitol.

(iii) Water - It is required to dissolve gelatin and plasticizers and to maintain softness. Eg. Purified water.

(iv) Colorants / opacifiers - Used to give color and protect light-sensitive Products. -

Eg. Titanium dioxide.

(v) Preservatives - Used to prevent microbial growth in the moist shell.

Eg. Methyl paraben, Propyl paraben.

(vi) Flavoring agents - Used in chewable softgels for better taste.

Eg. Mint flavor, fruit flavors.

Manufacturing of Soft Gelatin Capsules

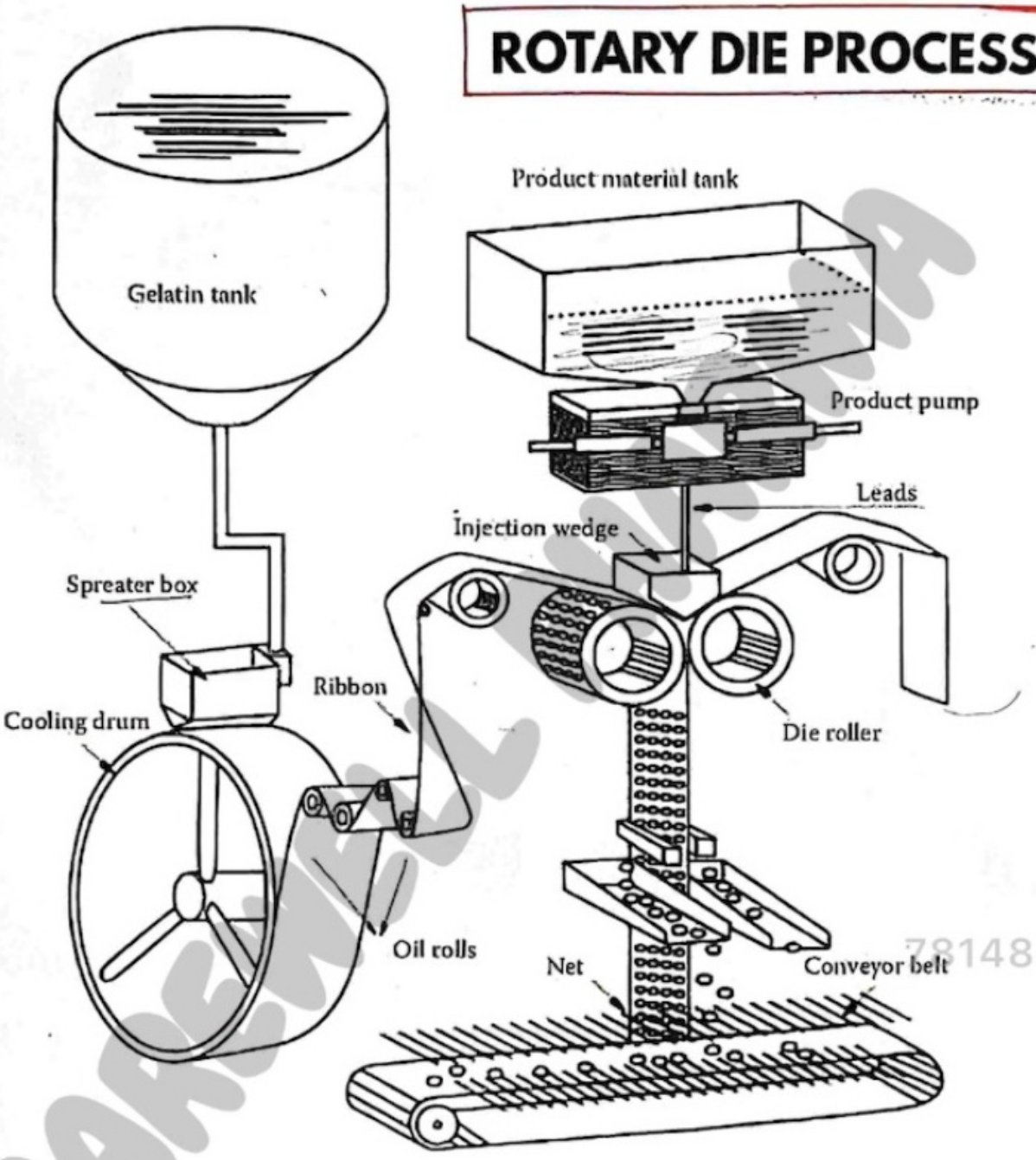
Soft gelatin capsules are manufactured through Rotary die process which includes several steps:-

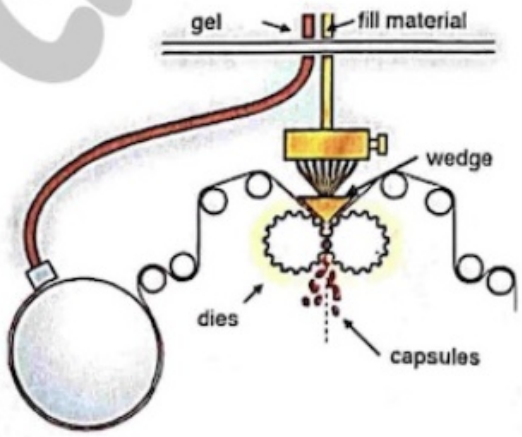
1. Preparation of gelatin mass. — By using formulation
2. Preparation of fill material — main drug / vitamins
3. Formation of ribbons.
4. Filling & sealing (Rotary die)
5. Drying / Curing.
6. Cleaning & Inspection.

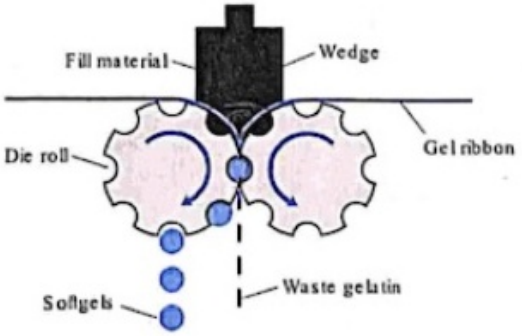


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ROTARY DIE PROCESS







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1. Preparation of Gelatin Mass -

The process begins by melting gelatin, water and plasticizers to make a soft, uniform and flexible gelatin mass for the capsule shell.

2. Preparation of Fill Material

At the same time, the drug is dissolved or dispersed in suitable oily or liquid base to prepare the fill material for the softgel.

3. Formation of Gelatin Ribbons.

The gelatin mass is spread into two wide, thin ribbons by cooling drums that roll the molten gelatin into soft sheets.

4. Filling and sealing (Rotary Die Process).

Both gelatin ribbons pass through rotating dies where the fill material is injected, and the two ribbons are pressed together to form, fill and seal the softgel capsule in one step.

5. Drying / Curing.

The sealed softgels are transferred to drying tunnels where controlled air removes excess moisture and gives the shell proper strength.

6. Cleaning and Inspection.

Finally, the dried softgels are cleaned, polished, inspected for defects, and then packed.

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Packaging, Storage & Stability Testing of Soft Gelatin Capsule.

(a) **Blister Packing** - Softgels are packed in (PVC) - Aluminum or PVDC blisters which protect each capsule separately from (moisture) and handling damage.
Eg. Vitamin E softgels in PVC - Aluminium blister.

(b) **Strip Packing** - (Two) layers of heat - sealed aluminium foil enclose individual softgels to give an airtight pack for moisture sensitive products.
Eg. Vitamin A softgels in aluminium - aluminium strip.

(c) **Bottle Packing (HDPE bottles)** - Softgels are filled in (plastic) bottles with cotton or cushioning to avoid sticking and breaking during transport.
Eg. Omega-3 softgel in HDPE bottles.

(d) **Use of Desiccants** - Silica (gel) or molecular sieves are placed in bottles to absorb extra (moisture) and keep softgels stable.
Eg. Silica gel (pouches) inside HDPE bottles.

(e) **Light Resistant Pack** - Amber bottles or (opaque) blisters are used for light-sensitive formulations like Vitamin A/E softgels.
Eg. Vitamin A and E softgels in amber HDPE bottles.

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Storage of Soft Gelatin Capsule

- Store at cool temperature (15-25°C) because heat softens the gelatin shell.
- Low humidity (around 35% RH) is required because softgels contains more water and stick together in high humidity.
- Keep away from direct sunlight to avoid degradation of sensitive vitamins and oils.
- Airtight bottles prevent moisture exchange and protect the shell from becoming too soft or too hard.

Applications.

- Used for Vitamin A, D, E, K, Omega-3 and fish oil.
- Softgels help dissolving poorly soluble drugs better, giving faster absorption.
- Ideal for drugs that cannot be made into tablet or hard capsule.
- Multiple oils, vitamins, or lipid-based drugs can be mixed together
- Soft, smooth shell makes swallowing easier and improves patient acceptance.

Stability Testing of Soft Gelatin Capsules

- Stability testing checks how soft gelatin capsules maintain their quality, safety and performance over time when stored at different temperature and humidity conditions.
- It helps to decide their shelf-life and expiry date.

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Parameters Tested

1. Physical Appearance - Check color change, sticking, leakage, softening or hardening of capsules.
2. Assay (Drug Content) - Tests if the drug amount remains within limit during storage.
3. Dissolution / Disintegration - checks if the capsule still release drug properly after months of storage.
4. Moisture Content - Measures changes in moisture because softgels absorb and lose water easily.
5. Impurity / Degradation Study - Check for new impurities or oxidation products (especially for oils / vitamins).
6. Microbial Stability - Ensure microbial growth does not increase because softgels contain water.

Quality Control Tests

1. Leakage / seal Integrity Test - This tests checks whether soft gelatin capsules are properly sealed and ensures that no liquid leaks out of the capsule.

Procedure -

1. Visual check - Capsules are inspected for any wet spots or oil leakage on the surface.
2. Dye tests - Capsules are dipped in dye solution, if dye enters the capsule, the seal is week.
3. Vaccum / pressure test - Weakly sealed capsules leak when exposed to vaccum or pressure.

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Result / Acceptance -

No capsule should show leakage, wetness or dye entry.

Other Test.

These tests are same as hard gelatin capsules.

- Appearance ~
- Weight variation ~
- Content uniformity ~
- Disintegration ~
- Dissolution ~
- Assay. ~

Already covered in Hard Gelatin Capsules

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Q.2 Explain the formulation, requirement and process of pelletization for pharmaceutical Pellets

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Pellets are small, free-flowing, spherical or semi-spherical solid units containing drug and excipients, usually with a size between 0.5 - 1.5 mm. ⬭ ◯ ▭ ← Pellets

- They are prepared by a process called pelletization, which converts powders into round, uniform particles for better flow, uniform release and improved patient compliance.

Eg. Pantoprazole enteric-coated pellets →



Advantages

- Improve flow and uniformity
- Reduce dose dumping.
- Allow controlled / sustained release
- Improve appearance and packing
- Allow taste-making.



Disadvantages

- Pellet production is expensive.
- Process needs multiple steps and takes more time.
- Require skilled operators and strict control.
- Pellets may break or stick if not dried properly.

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Properties:

- Uniform size (0.5 - 1.5 mm)
- Smooth spherical shape.
- Good mechanical strength.
- Low dust content
- Good flow property.
- Uniform drug distribution.

Formulation of Pellets

Pellets formulation usually contains:

1. Active Pharmaceutical Ingredients (API) - It is the (Main) drug
   - eg. Omeprazole, Diclofenac.
2. Diluent / filler - It Adds bulk and helps pellet formation.
   - eg. Lactose, Microcrystalline cellulose (MCC).
3. Binder - It helps particles (stick) and form pellets.
   - eg. PVP, HPMC, Starch paste.
4. Disintegrating agents - It Helps (pellets) break after ingestion.
   - Crospovidone, Sodium starch glycolate.
5. Coating Polymers (for release control) - It modify drug release (controlled, sustained, entric).
   - eg. Eudragit polymers, Ethylcellulose.
6. Plasticizer (in coating) - It makes coating flexible
   - eg. Triethyl citrate, PEG.
7. Solvent / Granulating fluid - It moistens powder for pellet formation
   - eg. water, Alcohol - water mix.

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Process of Pelletization.

It includes several steps :-

(i) Dry mixing.
(ii) Wet massing (Granulation).
(iii) Extrusion
(iv) Spheronization
(v) Drying.
(vi) Screening & polishing.

API

(i) Dry mixing - The process begins by mixing the drug with diluent, binder and other excipients to form a uniform dry powder blend.

(ii) Wet Massing (Granulation) - A suitable liquid (water or binder soln) is added to the dry mix to form a soft, plastic and workable wet mass suitable for extrusion.

solidness

(iii) Extrusion - The wet mass is pased through an extruder where it is forced through small cylindrical holes to produce long spaghetti-like strands of uniform thickness.

(extrudates)

(iv) Spheronization - The wet extruder are broken and rolled in a fast-rotating spheronizer plate, which rounds them into smooth, spherical pellets.

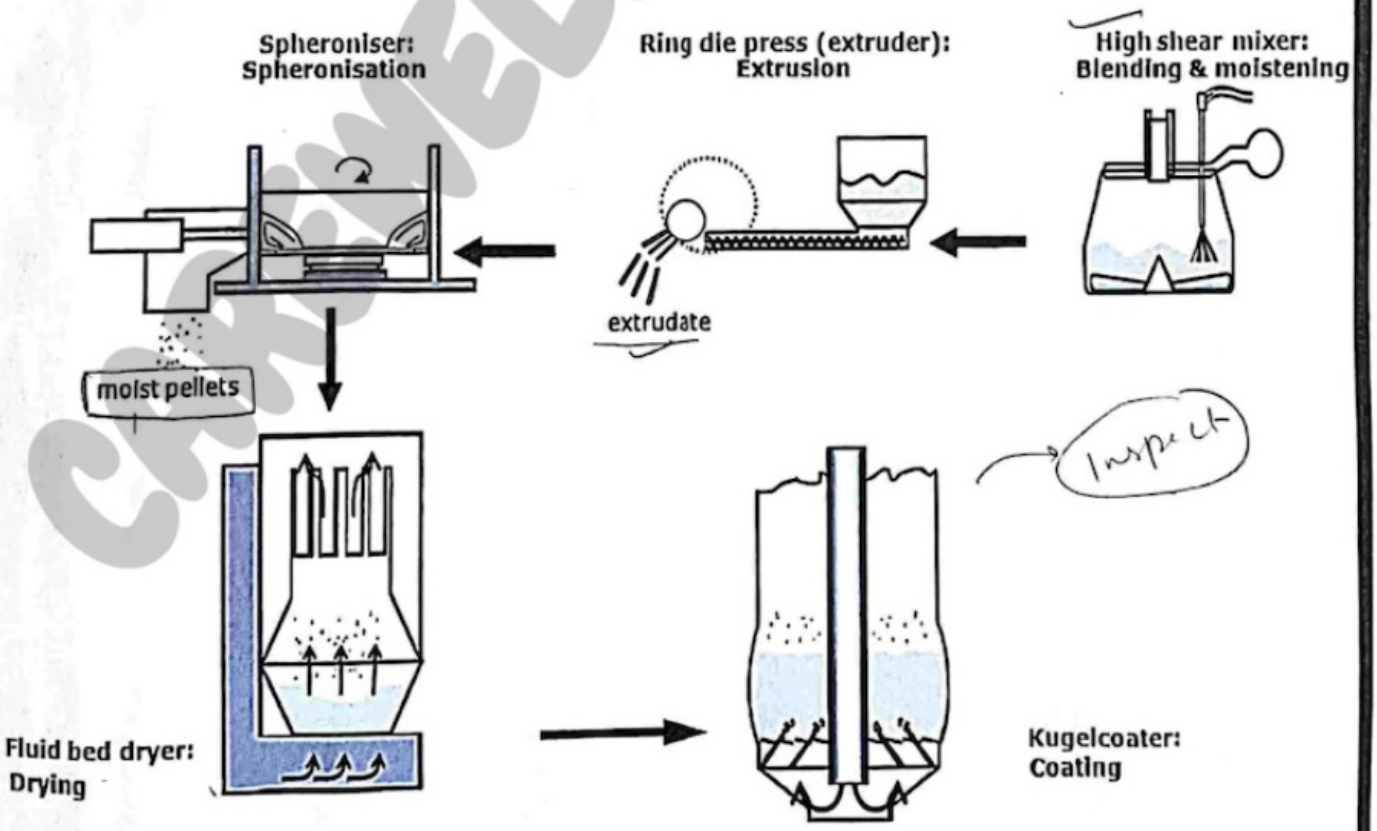
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(v) Drying - The freshly formed pellets are dried in a tray dryer or fluid bed dryer to remove moisture and achieve proper hardness.

(vi) Screening & Polishing - The dried pellets are passed through sieves to separate uniform - sized pellets and remove broken or oversized particles.

PELLETIZATION PROCESS



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